CCS4004 – Optimization Techniques

Practice Problem Set

Amgad A. Salama, Ph.D.

1. Optimization Types & Classification

Problem 1.1 Lemonade Stand Classification

A street vendor wants to maximize daily profit from selling two drinks: lemonade and iced tea. Each product has a fixed selling price and fixed production cost per cup. The vendor faces limits on daily sugar supply (in kg), available cups, and working hours. All quantities are continuous.

a) Identify the **decision variables**.

b) Identify the **objective function** (describe in words and as a mathematical expression).

c) List the **constraints** in plain English.

d) Classify this problem as LP, NLP, or IP and justify your answer.

Problem 1.2 Student Schedule Classification

A student can choose to enroll in at most 5 elective courses next semester. Each course has a known grade-point contribution and requires a certain number of weekly study hours. The student has a maximum of 20 study hours per week available for electives.

a) What type of variable must be used to model whether a course is taken or not? Why?

b) Write the objective function if the student wants to maximize total expected GPA contribution.

c) Write the study-hours constraint.

d) Classify this problem (LP, NLP, or IP) and explain what feature forces that classification.

Problem 1.3 WiFi Router Placement

An engineer wants to place a single WiFi router inside a rectangular apartment to maximize total signal coverage. Signal strength at any point decreases with the *square* of the distance from the router.

a) What are the decision variables?

b) Why can this **not** be classified as a Linear Program?

c) Classify this problem correctly and state the feature that determines the classification.

d) Give one real-world example from the slides that belongs to the *same* class as this problem.

Problem 1.4 Diet Planning vs. Production Mix

Consider the two scenarios below:

Scenario A – Diet Problem: A nutritionist must design a minimum-cost weekly meal plan. Available foods have known costs, calorie counts, protein, fat, and carbohydrate contents per serving. Daily minimum and maximum intake requirements exist for each nutrient. All serving sizes are continuous.

Scenario B – Factory Scheduling: A factory produces three products on two machines. Each product requires a whole number of machine-hours (cannot be split). Profits are known per unit. Machine capacity is limited per week.

a) Classify **Scenario A** (LP, NLP, or IP). Write a compact mathematical formulation.

b) Classify **Scenario B**. Write a compact mathematical formulation.

c) Explain the key structural difference between the two models.

d) If serving sizes in Scenario A must also be whole numbers of servings, how does the classification change?

Problem 1.5 Amazon Delivery Route

Amazon must assign each of 50 packages to one of 3 delivery trucks, and then determine the delivery sequence for each truck. Each truck starts and ends at a central depot. Travel time between any two locations is known. The objective is to minimize the total time all trucks spend on the road.

a) Define appropriate decision variables for the assignment of packages to trucks.

b) Define additional decision variables for sequencing deliveries within each truck.

c) Write the objective function.

d) Classify this problem. Is it LP, NLP, or IP? Justify carefully.

e) What is the well-known OR problem name for this type of scenario (mentioned in the slides)?

Problem 1.6 Portfolio Management

An investor allocates a budget B across n assets. Let x_i be the fraction of budget invested in asset i, μ_i the expected return of asset i, σ_{ij} the covariance between assets i and j, and ρ a risk-tolerance parameter.
The investor wants to maximize expected return minus a penalty for risk:

$$\max \quad \sum_{i=1}^{n} \mu_i x_i \; - \; \rho \sum_{i=1}^{n} \sum_{j=1}^{n} \sigma_{ij} x_i x_j$$

subject to: $\sum_{i=1}^{n} x_i = 1$ and $x_i \geq 0$ for all i.

a) Classify this problem. Is it LP, NLP, or IP?

b) Which specific term in the objective makes it belong to this class?

c) If the risk penalty term were removed, what would the problem become?

d) If each $x_i \in \{0, 1\}$ (invest fully or not at all), what class would it become?

Problem 1.7 Hospital Resource Allocation

A hospital operates three departments: Emergency (E), Surgery (S), and Outpatient (O). It must allocate n nurses across these departments for a single shift. Let x_E, x_S, x_O denote the number of nurses assigned to each department.

- Nurse effectiveness score per department: $e_E = 4$, $e_S = 7$, $e_O = 3$.
- Total nurses available: 30.
- Emergency requires at least 8 nurses; Surgery requires at least 5; Outpatient at least 3.
- The hospital also wants to ensure that nurses in Surgery are no more than twice the number in Emergency: $x_S \leq 2x_E$.
- Nurses must be whole numbers.

a) Formulate the complete mathematical model (variables, objective, all constraints).

b) Classify the problem type.

c) If the constraint $x_S \leq 2x_E$ is removed and all variables are relaxed to be continuous, which type does the problem reduce to?

d) Explain why this rounding from the relaxed solution is not always valid for IP problems.

e) Add a new binary variable y to model the following policy: *"If Surgery receives more than 10 nurses, Outpatient must receive at least 6."* Write the constraint(s) that enforce this using y.

Problem 1.8 Multi-Objective Production

A factory produces two products, A and B. The production quantities are x_A and x_B (continuous, ≥ 0). Known data:

	Profit (\$/unit)	Labor (hr/unit)	Machine (hr/unit)	CO_2 (kg/unit)
Product A	5	2	1	3
Product B	4	1	2	1
Available	—	100 hr	80 hr	$\leq$ 200 kg

a) Formulate and classify the profit-maximization LP.

b) Now suppose the factory wants to *simultaneously* maximize profit and minimize CO_2 emissions. Write both objectives and explain why this is now a multi-objective problem.

c) If CO_2 is converted to a constraint (emission $\leq$ 200 kg), does the problem remain LP? Justify.

d) Suppose machine processing rate depends on temperature, giving a quadratic machine-time term $0.01x_A^2$ added to the machine constraint. Reclassify the problem and explain.

Problem 1.9 Blending and Classification Boundaries

An oil refinery blends three crude oils (C1, C2, C3) into two fuel products (F1, F2). Let x_{ij} denote the volume (barrels) of crude i used in fuel j.

- Profit per barrel: F1 gives \$12/barrel, F2 gives \$9/barrel.
- Supply of crude: C1: 200 bbl, C2: 300 bbl, C3: 250 bbl.
- Quality constraint for F1: the fraction of C1 in F1 must be $\geq 40\%$.
- Quality constraint for F2: the fraction of C3 in F2 must be $\leq 30\%$.
- All volumes are continuous.

a) Define all decision variables and write the objective function.

b) Show that the quality constraints can be written as *linear* inequalities (hint: multiply both sides by the total volume of each fuel).

c) Write the complete supply constraints.

d) Classify the resulting model and justify.

e) If the refinery must decide whether to *activate* each blending stream ($x_{ij} > 0$ triggers a fixed setup cost), introduce binary variables and explain how the classification changes.

2. Problem Formulation: Verbal Description to Mathematical Model

Problem 2.1 Bakery Production

A bakery makes cakes and cookies. Each cake earns \$8 profit and uses 2 kg of flour and 1 hour of oven time. Each batch of cookies earns \$5 profit and uses 1 kg of flour and 0.5 hours of oven time. The bakery has 20 kg of flour and 10 hours of oven time per day.
Formulate the complete LP model to maximize daily profit:

a) Define decision variables clearly.

b) Write the objective function.

c) Write all constraints including non-negativity.

d) Identify what type of constraint each one represents (resource, demand, logical).

Problem 2.2 Laptop Bag Packing

A student packs a bag for university with a weight limit of 5 kg. Items available and their properties are:

Item	Weight (kg)	Usefulness Score	Take at most
Laptop	2.0	10	1
Textbook	1.5	6	2
Water bottle	0.5	4	1
Notebook	0.3	3	3

a) Define decision variables.

b) Write the objective (maximize total usefulness).

c) Write all constraints.

d) Classify the model type and justify.

Problem 2.3 Crop Irrigation

A farmer grows wheat and corn on a 100-hectare farm. Each hectare of wheat requires 500 litres of water and yields \$200 profit. Each hectare of corn requires 800 litres of water and yields \$300 profit. The farmer has 60,000 litres of water available. At least 20 hectares must be planted with wheat due to a contractual obligation.

a) Define decision variables.

b) Write the objective function.

c) List and write all constraints with justification for each.

d) Classify the model.

Problem 2.4 Furniture Factory

A furniture factory produces chairs, tables, and shelves. Resource requirements and profits are summarized below:

Product	Profit (\$)	Wood (m^2)	Labor (hr)	Paint (L)
Chair	40	3	2	1
Table	90	8	4	2
Shelf	30	2	1	0.5
Available	—	240	100	50

Additionally, the factory must produce *at least* 5 tables per week due to a standing order, and cannot produce more than 20 chairs (storage constraint).

a) Define all decision variables.

b) Write the complete LP formulation.

c) Identify which constraints are resource constraints, lower-bound constraints, and upper-bound constraints.

d) If the factory could only produce whole-number quantities, how does the model change?

Problem 2.5 Nurse Shift Scheduling

A hospital needs to cover three 8-hour shifts (Morning: 6am–2pm, Afternoon: 2pm–10pm, Night: 10pm–6am) over one day. Minimum nurse requirements per shift are 10, 8, and 6 respectively. Nurses starting the morning shift earn \$120/shift; afternoon \$130/shift; night \$160/shift. The hospital wants to minimize total daily wage cost.

a) Define decision variables.

b) Write the objective function.

c) Write the demand constraints.

d) Write non-negativity and integrality constraints.

e) Classify the complete model.

f) If nurses are allowed to work consecutive shifts (8+8 = 16 hours), how would you extend the model? Describe the additional variables and constraints needed.

Problem 2.6 Advertising Budget Allocation

A company has an advertising budget of \$50,000 to allocate across four channels: TV (x_1), Social Media (x_2), Print (x_3), and Radio (x_4). Expected customer reach per \$1,000 spent: 500, 800, 200, 150 respectively.
Additional requirements:

- TV spending must be at least \$10,000.
- Social Media cannot exceed 40% of the total budget.
- Print + Radio combined must be at least \$5,000.
- All allocations in multiples of \$1,000 (integer multiples).

a) Formulate the complete model (maximize total reach).

b) Write the "Social Media $\leq$ 40%" constraint as a standard linear inequality.

c) Classify the model. Would relaxing the integrality change the class?

d) Suppose TV reach has diminishing returns: reach $= 500x_1 - 0.01x_1^2$. How does this affect the model classification?

Problem 2.7 Multi-Period Production Planning

A factory plans production over 3 months. Demand for its single product is: Month 1: 100 units, Month 2: 150 units, Month 3: 120 units. Let x_t = units produced in month t, I_t = inventory at end of month t.

- Production cost: \$10/unit in Month 1, \$12/unit in Month 2, \$11/unit in Month 3.
- Holding cost: \$2/unit/month.
- Initial inventory $I_0 = 20$ units.
- Maximum production capacity per month: 160 units.
- Final inventory $I_3 \geq 10$ units (safety stock).

a) Define all decision variables with their meaning.

b) Write inventory balance constraints for each month: $I_t = I_{t-1} + x_t - d_t$ where d_t is demand in month t.

c) Write the complete objective function (minimize total production + holding cost).

d) Write all constraints (capacity, non-negativity, safety stock).

e) Classify the model. Could this be solved as a simple LP?

Problem 2.8 Facility Location with Binary Decisions

A logistics company is considering opening up to 3 distribution centres (DC1, DC2, DC3) to serve 4 customer zones (Z1, Z2, Z3, Z4). Data:

	Z1	Z2	Z3	Z4	Fixed Cost
DC1	\$4	\$8	\$6	\$9	\$1000
DC2	\$7	\$3	\$5	\$6	\$900
DC3	\$5	\$6	\$4	\$3	\$800

(Shipping costs are per unit, demand per zone: 50, 70, 60, 80 units.)

a) Define binary variables for opening DCs and assignment variables.

b) Write the objective function (minimize total fixed + shipping cost).

c) Write the constraint: "each zone must be served by exactly one DC."

d) Write the constraint: "a zone can only be assigned to an open DC."

e) Write the constraint: "at most 2 DCs can be opened."

f) Classify the model and explain why this cannot be solved as an LP.

Problem 2.9 Project Selection under Budget and Precedence

A company evaluates 5 investment projects ($P_1, \ldots, P_5$) with the following data:

Project	NPV (\$K)	Cost (\$K)	Requires
P_1	120	50	—
P_2	80	40	P_1 must be selected
P_3	150	70	—
P_4	60	30	P_3 must be selected
P_5	200	90	P_1 and P_3 both selected
Total budget:		\$200K	

a) Define binary decision variables.

b) Write the objective function (maximize total NPV).

c) Write the budget constraint.

d) Write the *precedence* constraints for P_2, P_4, and P_5 using binary variables.

e) Add a constraint: "at most 3 projects can be selected."

f) Classify the problem and explain what solution method class applies.

3. OR Model Families: Recognition and Formulation

Problem 3.1 Recognize the Model

For each scenario below, identify the most appropriate **OR model family** (allocation, transportation, assignment, network flow, inventory, queuing, simulation). Briefly justify each answer.

a) A university assigns 30 teaching assistants to 30 lab sections, each TA to exactly one section, minimizing the sum of student-dissatisfaction scores.

b) A supermarket manager wants to know the average customer waiting time at checkout, given random arrival and service rates.

c) A city's water authority wants to find the minimum-cost pipe network design to connect 8 districts.

d) A clothing brand has 4 factories and 6 retail stores; it wants to ship garments from factories to stores at minimum total freight cost.

e) A pharmacy decides how many units of a vaccine to order per batch to minimize ordering and refrigerated storage costs.

Problem 3.2 Transportation Model Basics

Two warehouses (W1, W2) supply three retail stores (R1, R2, R3). Supplies: W1 = 80 units, W2 = 100 units. Demands: R1 = 50, R2 = 70, R3 = 60 units. Shipping costs (per unit):

	R1	R2	R3
W1	\$2	\$3	\$5
W2	\$4	\$1	\$2

a) Is supply equal to demand? If not, how is the model usually handled?

b) Define decision variables x_{ij}.

c) Write the objective function.

d) Write all supply and demand constraints.

Problem 3.3 Assignment Model Basics

A software company has 3 developers (Alice, Bob, Carol) and 3 projects (P1, P2, P3). Each developer must be assigned to exactly one project. Estimated completion times (days):

	P1	P2	P3
Alice	6	8	5
Bob	7	4	9
Carol	5	6	7

a) Define binary decision variables.

b) Write the objective function (minimize total completion time).

c) Write the assignment constraints (one project per developer, one developer per project).

d) Explain why this is a special case of the transportation model.

Problem 3.4 Multi-Source Transportation with Capacity

Three factories (F1, F2, F3) ship to four distribution centres (D1, D2, D3, D4). Supplies: 200, 180, 220 units. Demands: 120, 150, 180, 100 units (balanced after adding a dummy if needed).
Additional constraints:

- Road capacity limits: $x_{12} \leq 80$ (F1→D2), $x_{31} \leq 60$ (F3→D1).
- F2 cannot ship to D3 (route unavailable): $x_{23} = 0$.

Shipping costs (c_{ij}) are given in the cost matrix below:

$$C = \begin{pmatrix} 3 & 7 & 4 & 6 \\ 5 & 2 & \infty & 3 \\ 4 & 6 & 5 & 2 \end{pmatrix}$$

a) Is the problem balanced? If not, add a dummy source or destination.

b) Write the complete model including capacity and route constraints.

c) How does the "∞" cost for x_{23} enforce the route-unavailable constraint?

d) If F2→D3 could be opened for a fixed cost of \$500, how would you model this with a binary variable?

Problem 3.5 Crew Assignment in an Airline

An airline has 4 crews (C1–C4) available for 4 flights (F1–F4) on a given day. Each crew has received different training ratings (0–10) for each flight type. Ratings matrix:

$$R = \begin{pmatrix} 8 & 5 & 7 & 6 \\ 4 & 9 & 5 & 8 \\ 7 & 6 & 8 & 4 \\ 5 & 7 & 6 & 9 \end{pmatrix}$$

where R_{ij} = rating of crew i for flight j. Each crew is assigned to exactly one flight and each flight has exactly one crew.

a) Write the complete assignment model (maximize total rating).

b) Suppose crew C3 is unavailable for Flight F2 due to a rest-hour violation. How do you handle this? Modify the model.

c) Show how this assignment problem is a special case of the transportation model by rewriting it in transportation form.

d) If the airline wants to maximize *minimum* rating across all assignments (a max-min objective), explain why this is no longer a standard assignment problem and suggest how to reformulate it using an auxiliary variable.

Problem 3.6 Stochastic Inventory vs. Deterministic EOQ

A retailer sells 1,000 units/year of a product. Ordering cost = \$200/order. Holding cost = \$4/unit/year.

a) Compute the EOQ (Q^*) using the formula from the slides.

b) Compute the number of orders per year and the total annual cost.

c) If the lead time is 2 weeks and demand is deterministic, compute the reorder point.

d) Now suppose demand is random with mean 1,000 and standard deviation 100 units/year. Explain (without full derivation) how this changes the model and what additional cost term appears.

e) What type of OR model (deterministic vs. stochastic, static vs. dynamic) does each case (c) and (d) represent?

Problem 3.7 Multi-Commodity Transportation

A company ships two products (A and B) from 2 plants to 3 warehouses. Each arc (i, j) has a *shared capacity* u_{ij} (total units of both products combined). Let x_{ij}^A and x_{ij}^B be flows of each product on arc (i, j). Supply: Plant 1 has 100A, 80B; Plant 2 has 120A, 60B. Demand: WH1 needs 70A, 50B; WH2 needs 90A, 40B; WH3 needs 60A, 50B. Arc capacities: $u_{11} = 80$, $u_{12} = 90$, $u_{13} = 70$, $u_{21} = 60$, $u_{22} = 80$, $u_{23} = 90$.

a) Write the supply constraints for each product at each plant.

b) Write the demand constraints for each product at each warehouse.

c) Write the shared capacity constraint for each arc.

d) Write the objective (minimize total shipping cost with given cost matrices C^A and C^B).

e) Explain why single-commodity formulation cannot directly handle this problem.

Problem 3.8 Warehouse Location: Full MIP

Extend the warehouse location model from the slides. A company considers 3 potential warehouses (W_1, W_2, W_3) and 5 customer zones $(C_1, \ldots, C_5)$.

- Fixed opening costs: \$5000, \$4000, \$6000.
- Each warehouse has a capacity of 300 units/year.
- Customer demands: 80, 100, 90, 70, 110 units/year.
- Unit shipping costs c_{ij} are given in the matrix:

$$C = \begin{pmatrix} 2 & 5 & 3 & 6 & 4 \\ 4 & 2 & 6 & 3 & 5 \\ 3 & 4 & 2 & 5 & 6 \end{pmatrix}$$

a) Define all binary and continuous decision variables.

b) Write the complete objective function.

c) Write customer assignment constraints.

d) Write the warehouse capacity constraints.

e) Write the "link" constraints: a customer can only use an open warehouse.

f) Write the constraint: "at least 2 warehouses must be opened."

g) Classify and explain what makes this harder to solve than a pure LP.

Problem 3.9 Dynamic Inventory Over 4 Periods

A company manages inventory of a single product over 4 periods. Demand: $d_1 = 40$, $d_2 = 60$, $d_3 = 50$, $d_4 = 70$ units. A fixed ordering cost $K = \$300$ is incurred whenever an order is placed. Variable ordering cost: \$5/unit. Holding cost: \$2/unit/period. Initial inventory $I_0 = 10$. Final inventory must be ≥ 0.
Let $y_t \in \{0, 1\}$ indicate whether an order is placed in period t, $q_t \geq 0$ be the order quantity, and $I_t \geq 0$ be ending inventory.

a) Write the objective function.

b) Write inventory balance constraints for each period.

c) Write the constraint that links y_t and q_t: an order can only be placed if $y_t = 1$ (use a big-M constraint, with $M = \sum d_t$).

d) Write all non-negativity and binary constraints.

e) Classify the complete model.

f) Explain why this is called a *dynamic* OR model and contrast it with the static EOQ model.

4. Network Flow Models

Problem 4.1 Shortest Path by Inspection

Consider the network below (same structure as the slides example). Arc costs: A→B=4, A→C=2, B→D=3, C→E=3, B→C=1, D→F=2, E→F=4, C→D=2.

a) List **all** simple paths from A to F.

b) Compute the total cost of each path.

c) Identify the shortest path and its cost.

d) If arc B→D is blocked (e.g., road closure), which path is the new shortest?

Problem 4.2 Shortest Path LP Formulation

Using the same network as Problem 4.1, formulate the shortest path problem as a **linear program** using flow variables $f_{ij} \in \{0, 1\}$ on each arc.

a) Define the flow variables.

b) Write the objective function (minimize total arc cost).

c) Write the flow balance constraints at each node: (outflow − inflow = supply/demand, where source A has supply 1, sink F has demand 1).

d) Write the variable bounds.

e) How many decision variables and constraints does this LP have?

Problem 4.3 Maximum Flow Problem

A water supply network has a source (S), a sink (T), and 4 intermediate nodes. Arc capacities (litres/hour):

Arc	Capacity	Arc	Capacity
S→A	15	B→T	10
S→B	10	C→T	12
A→C	12	D→T	8
A→D	8	B→D	5

a) Formulate the max-flow LP (maximize flow out of S, subject to capacity and conservation constraints at all intermediate nodes).

b) By inspection or enumeration, find the maximum feasible flow value.

c) Identify a minimum cut and verify that its capacity equals the max flow (state the Max-Flow Min-Cut theorem).

Problem 4.4 Minimum Cost Flow

A logistics company routes goods through a network of 5 nodes. Node 1 is a supplier (supply = 30), Node 5 is a customer (demand = 30), Nodes 2, 3, 4 are transshipment nodes (supply = demand = 0).

Arc	Capacity	Unit Cost
1→2	20	\$3
1→3	25	\$2
2→4	15	\$4
3→4	20	\$1
3→5	15	\$5
4→5	25	\$2

a) Formulate the minimum cost flow LP (variables, objective, flow balance, capacity).

b) How many feasible paths exist from Node 1 to Node 5?

c) By enumeration or reasoning, find the optimal routing plan.

d) If the capacity of arc 3→4 is reduced to 10, how does the optimal solution change?

Problem 4.5 Project Scheduling with CPM Network

A construction project has 6 activities (A–F) with durations (days): A=3, B=5, C=4, D=2, E=6, F=3. Precedence: A before C, A before D; B before D; C before E; D before F; E and F before finish.

a) Draw the activity-on-arc (AOA) network.

b) Compute Early Start (ES) and Late Start (LS) times for each activity.

c) Identify the critical path and project duration.

d) Formulate the project duration minimization as a **shortest path problem** on the AOA network. What is being "minimized"?

e) If activity E can be crashed (shortened) by up to 2 days at a cost of \$500/day, write the LP that minimizes crashing cost while keeping total project duration ≤ 13 days.

Problem 4.6 Telecommunications Network Design

A telecom company designs a network connecting 5 cities. It must select which links to install (from 8 candidate links) so that every city is reachable from every other city (a connected spanning network), while minimizing total installation cost.

Link	Cities Connected	Cost ($K)
L1	1–2	10
L2	1–3	15
L3	2–3	8
L4	2–4	12
L5	3–4	9
L6	3–5	14
L7	4–5	7
L8	1–5	20

a) Which OR network problem does this correspond to?

b) Using Kruskal's or Prim's algorithm (your choice), identify the minimum spanning tree.

c) What is the minimum total installation cost?

d) If the company must also ensure that at least 2 link-disjoint paths exist between cities 1 and 5, how does the problem complexity change?

Problem 4.7 Multi-Period Distribution Network

A distributor serves 3 retailers over 2 time periods. In each period they can ship from a central depot or from inter-period stored inventory.

- Period 1 depot supply: 200 units; Period 2 depot supply: 180 units.
- Retailer demands by period: R1: (60, 70), R2: (80, 90), R3: (50, 60).
- Shipping cost depot→retailer per period: $3, $4, $5 (for R1, R2, R3).
- Inventory can be held at the depot between periods: $1/unit.
- Each shipment arc has capacity 100 units.

a) Model this as a **time-expanded network**: define nodes for each (location, period) pair and draw the arc structure.

b) Define all decision variables.

c) Write the complete LP (flow balance at each node-period, capacity, non-negativity).

d) Write the objective function.

e) Explain how the time-expanded network converts a dynamic problem into a static network flow.

Problem 4.8 Capacitated Vehicle Routing

A depot serves 5 customers using 2 identical trucks, each with capacity $Q = 60$ units. Customer demands: $d_1 = 15, d_2 = 25, d_3 = 20, d_4 = 30, d_5 = 10$. Travel cost matrix (symmetric, in \$):

$$C = \begin{pmatrix} 0 & 4 & 6 & 5 & 8 & 3 \\ 4 & 0 & 3 & 7 & 5 & 9 \\ 6 & 3 & 0 & 4 & 6 & 7 \\ 5 & 7 & 4 & 0 & 3 & 8 \\ 8 & 5 & 6 & 3 & 0 & 5 \\ 3 & 9 & 7 & 8 & 5 & 0 \end{pmatrix}$$

(Row/column 0 = depot, rows/columns 1–5 = customers.)

a) Let $x_{ijk} = 1$ if truck k travels arc (i, j). Write the objective.

b) Write the constraint: each customer is visited exactly once by exactly one truck.

c) Write the truck capacity constraint.

d) Write the flow conservation constraint for each truck at each node.

e) What is the name of this classic OR problem? How does it relate to the Traveling Salesman Problem (TSP)?

Problem 4.9 Network Flow with Side Constraints

Consider a min-cost flow network with 6 nodes and the following arc set, costs, and capacities:
Arcs: (1,2,c=3,u=50), (1,3,c=5,u=40), (2,4,c=2,u=30), (3,4,c=4,u=35), (2,5,c=6,u=20), (3,6,c=3,u=25), (4,6,c=2,u=40), (5,6,c=4,u=30)

Supply: node 1 = 60; Demand: node 6 = 60; all other nodes: balanced (0).
Additional side constraints:

- Total flow through arc $(2, 4)$ plus arc $(3, 4)$ must be ≤ 45.
- If flow on arc $(1, 2) \geq 30$, then flow on arc $(1, 3)$ must be ≥ 15.

a) Write the baseline min-cost flow LP (ignoring side constraints).

b) Write the first side constraint as a linear inequality.

c) For the second side constraint (conditional), introduce a binary variable z and write equivalent linear constraints using big-M.

d) Explain why adding the binary variable changes the problem class.

e) If both side constraints are removed, is an LP-optimal solution always integer for pure network flow problems? State the relevant theorem.

5. Inventory Models

Problem 5.1 Standard EOQ

A bookstore sells 2,400 textbooks per year. The cost to place an order is \$80. The holding cost is \$6 per book per year.

a) Compute the Economic Order Quantity Q^*.

b) Compute the number of orders placed per year.

c) Compute the time between orders (in weeks, assuming 52 working weeks/year).

d) Compute the total annual inventory cost (holding + ordering).

e) If the ordering cost doubles to \$160, how does Q^* change? Explain the relationship.

Problem 5.2 EOQ with Lead Time

A pharmacy sells 1,800 units/year of a medication. Ordering cost = \$150/order. Holding cost = \$9/unit/year. Lead time for delivery = 3 weeks (assume 52 weeks/year, constant demand).

a) Compute Q^*.

b) Compute the reorder point (ROP): the inventory level at which a new order should be placed.

c) Compute the cycle time (time between orders) in weeks.

d) Sketch the inventory-level-over-time graph and label Q^*, ROP, lead time, and cycle time.

e) If demand increases to 2,400 units/year (all else equal), by how much does Q^* change (in percentage)?

Problem 5.3 EOQ Sensitivity Analysis

A hardware store orders bolts. Annual demand $D = 5,000$ units. Ordering cost $S = \$100$. Holding cost $H = \$2$/unit/year.

a) Compute Q^* and total annual cost TC^*.

b) If H increases by 20% (to \$2.40), compute the new Q^* and TC^*.

c) If S decreases by 30%, compute the new Q^* and TC^*.

d) In general, if S is multiplied by a factor k, by what factor does Q^* change?

e) Based on your answers, which parameter (S or H) has a larger impact on Q^*?

Problem 5.4 EOQ with Quantity Discounts

A retailer purchases items at a unit price that depends on order size:

Order Quantity	Unit Price
$0 \leq Q < 500$	\$10.00
$500 \leq Q < 1000$	\$9.50
$Q \geq 1000$	\$9.00

Annual demand $D = 2,000$ units. Ordering cost $S = \$200$. Holding cost $= 20\%$ of unit purchase price per year.

a) Compute Q^* for each price tier (using $H = 0.20 \times \text{price}$).

b) For each tier, check whether Q^* is within the valid range; if not, use the minimum valid quantity.

c) Compute total annual cost (purchase + ordering + holding) for each feasible option.

d) Identify the optimal order quantity and explain the discount trade-off.

Problem 5.5 EOQ with Backorders Allowed

A wholesaler allows backorders (unfulfilled demand is met in the next cycle). The model parameters are: $D = 1,200$ units/year, $S = \$250$ per order, $H = \$8$ per unit per year, $B = \$20$ per unit per year (backorder penalty cost).
The EOQ with backorders gives:

$$Q^* = \sqrt{\frac{2DS}{H}} \cdot \sqrt{\frac{H+B}{B}}, \qquad S^* = Q^* \cdot \frac{H}{H+B}$$

where S^* is the maximum backorder level.

a) Compute Q^* and S^*.

b) Compute the maximum inventory level $(Q^* - S^*)$.

c) Compute the total annual cost including backorder penalty.

d) Compare total cost with and without backorders. When is allowing backorders beneficial?

e) As $B \to \infty$, what happens to Q^* and S^*? Explain intuitively.

Problem 5.6 Multi-Item Inventory with Shared Budget

A store manages inventory of 3 items with the following data:

Item	Demand (units/yr)	Order Cost ($)	Holding Cost ($/unit/yr)	Unit Price ($)
A	3,000	150	5	20
B	1,500	100	8	35
C	2,000	200	6	25

The total purchase budget for a single order cycle is limited to $15,000.

a) Compute the unconstrained EOQ for each item.

b) Compute the total purchase cost per cycle at unconstrained EOQs.

c) If the budget constraint is violated, explain the Lagrangian approach to adjust order quantities (concept, no full derivation needed).

d) Alternatively, set up this problem as an LP/NLP. What is the objective? What are the decision variables and constraints?

e) Classify the resulting constrained model.

Problem 5.7 Economic Production Quantity (EPQ)

A manufacturer produces a product at a rate of $P = 8,000$ units/year while demand is $D = 5,000$ units/year. Setup cost per production run = $400. Holding cost = $10/unit/year.
The Economic Production Quantity is:

$$Q^*_{EPQ} = \sqrt{\frac{2DS}{H\left(1 - \frac{D}{P}\right)}}$$

a) Compute Q^*_{EPQ}.

b) Compute the maximum inventory level reached during a production run.

c) Compute the total annual cost.

d) Compare Q^*_{EPQ} with standard EOQ (same D, S, H). Which is larger and why?

e) If the production rate P increases to infinity, show that the EPQ formula reduces to the standard EOQ formula. What does this mean practically?

f) Write the EPQ model as a continuous optimization problem: minimize total cost function $TC(Q)$ and show how Q^*_{EPQ} is derived using calculus ($dTC/dQ = 0$).

Problem 5.8 (s, S) Inventory Policy Formulation

A retailer uses an (s, S) inventory policy: whenever inventory falls to or below reorder point s, an order is placed to raise inventory to level S. Annual demand $D = 2,400$ units. Ordering cost $K = \$300$. Holding cost $h = \$5$/unit/year. Demand is random: weekly demand $\sim$ Normal($\mu = 50, \sigma = 10$) (assume 48 weeks/year). Lead time = 2 weeks.

a) Define the decision variables s and S clearly.

b) Write the expected total cost function as a function of s and S (include expected holding, ordering, and safety stock costs conceptually).

c) Compute the reorder point s to achieve a 95% service level (use $z_{0.95} = 1.645$; demand during lead time: mean $= 2\mu$, std $= \sqrt{2}\,\sigma$).

d) Compute the optimal order-up-to level $S = s + Q^*$ where Q^* is the EOQ.

e) What type of OR model (static vs. dynamic, deterministic vs. stochastic) is the (s,S) policy? Justify.

Problem 5.9 Inventory Optimization as a Full LP/IP

A supply chain manager controls inventory over 5 periods. Each period has demand d_t (known): 30, 50, 40, 60, 35 units.

- Ordering cost: \$200 fixed if an order is placed + \$8 variable per unit.
- Holding cost: \$3/unit/period.
- Maximum order quantity per period: 120 units.
- Initial inventory: 10 units. Final inventory: ≥ 0.
- Shortage not allowed ($I_t \geq 0$).

a) Define decision variables: $y_t \in \{0, 1\}$ (order or not), $q_t \geq 0$ (order qty), $I_t \geq 0$ (ending inventory).

b) Write the complete objective function.

c) Write inventory balance constraints.

d) Write the big-M linking constraint between y_t and q_t.

e) Write all remaining constraints.

f) Classify the model and state its solution method.

g) What is the total number of decision variables and constraints?